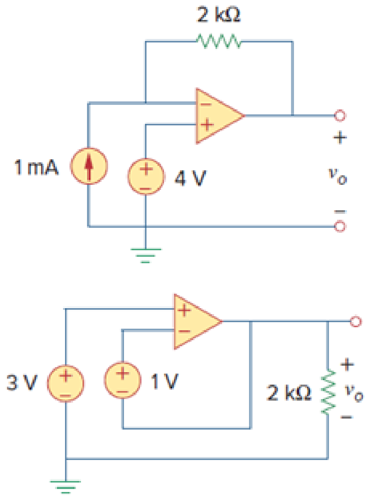
Chapter 5, Problem 9P

Problem

Determine v_o for each of the op amp circuits in Fig. 5.48.

Figure. 5.48:



Step-by-step solution

Step 1 of 4

The following is the given circuit diagram:

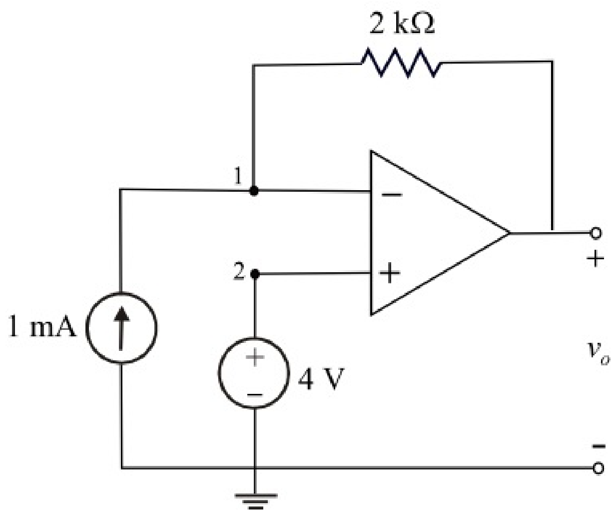


Figure 1

Step 2 of 4

The voltage at non inverting terminal is,

$v_2 = 4\text{ V}$

Since the given op amp is ideal op amp. Hence,

$v_1 = v_2 = 4\text{ V}$

Apply Kirchhoff's current law at node 1.

$$\frac{v_1 - v_o}{2\text{ k}} - 1\times10^{-3} = 0$$
$$\frac{v_1 - v_o}{2\text{ k}} = 1\times10^{-3}$$
$$v_1 - v_o = 2$$
$$\begin{aligned} v_o &= v_1 - 2 \\ &= 4-2 \\ &= 2\text{ V} \end{aligned}$$

Therefore, the output voltage, v_o is $\boxed{2\text{ V}}$.

Step 3 of 4

The following is the given circuit diagram:

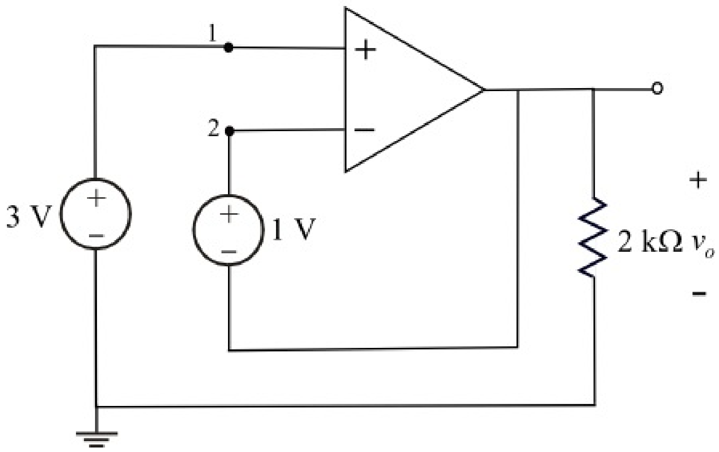


Figure 2

Step 4 of 4

From the Figure 2, the non-inverting input voltage is,

$v_1 = 3\text{ V}$

The inverting input voltage and non-inverting input voltages are same because the given op amp is ideal.

That is, $v_1 = v_2 = 3\text{ V}$

Apply Kirchhoff's voltage law to the Figure 2.

$v_2 - v_o = 1\text{ V}$ ...... (1)

Substitute equation $v_2 = 3\text{ V}$ in equation (1).

$$3 - v_o = 1$$
$$v_o = 3-1$$
$$v_o = 2\text{ V}$$

Therefore, the output voltage, v_o is $\boxed{2\text{ V}}$.